

Trajectory Variance: An Unsupervised Measure of Developmental Vocal Plasticity

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Abstract

How much does a vocalization change over the course of development? We propose trajectory variance, a per-vocalization plasticity score that answers this question without type labels. A displacement model learns to predict age-conditioned shifts in autoencoder latent space; the variance of its predictions across target ages quantifies how much each vocalization would change if produced at different developmental stages. Evaluated on three zebra finches (183K–274K vocalizations, 40–100 days post-hatch), trajectory variance separates learned song syllables from innate calls (Cohen’s $d = 0.29$ – 0.57 , $AUC = 0.58$ – 0.67 , after controlling for duration), while no nonparametric baseline achieves consistent separation. Acoustic correlations vary across birds—strong for one (spectral flatness $r = -0.56$), negligible for others—indicating that the measure captures developmental dynamics not reducible to static spectral form.

Index Terms: vocal development, birdsong, counterfactual generation, trajectory variance, optimal transport

1. Introduction

How much does a vocalization change over the course of development? Static acoustic descriptors—spectral shape, duration, energy—characterize what a sound is at the moment it is produced, but not how it evolves. Analogous questions arise in single-cell genomics, where optimal-transport methods track how cell states shift along developmental trajectories [1]. We propose **trajectory variance**, a measure that applies counterfactual reasoning to vocal development: if this vocalization had been produced at a different age, how different would it be?

The approach learns a model that predicts age-conditioned shifts in a latent representation of vocalizations. For each vocalization, counterfactual predictions are generated at several target ages, and the variance of these predictions serves as a plasticity score. Vocalizations predicted to change substantially across ages receive high trajectory variance; those predicted to remain stable receive low variance. No vocalization-type labels or manual annotation are needed.

We validate this approach on zebra finch vocal development—a well-studied model system where learned song syllables coexist with innate calls [2, 3]. During a critical period from roughly 25 to 90 days post-hatch (dph), juvenile birds progress from unstructured subsong through plastic song to a crystallized adult motif [4]. Individual syllables are shaped by local exploration [5] and gradually stabilized [6], while innate calls remain comparatively unchanged. Existing computational approaches classify vocalizations by their static acoustic form [7, 8, 9, 10, 11], or characterize developmental dynamics through repertoire dating [12] and generative modeling [13]. Trajectory variance complements

these: rather than classifying what a sound is or when similar sounds were produced, it measures how much a vocalization is predicted to change.

Applied to three birds (183K–274K vocalizations each, 40–100 dph), trajectory variance from the displacement model reliably separates song from calls with small-to-medium effect sizes (Cohen’s $d = 0.29$ – 0.57 , $AUC = 0.58$ – 0.67 after controlling for duration), while no nonparametric baseline—Gaussian OT, per-age k -NN, or per-age optimal assignment—achieves consistent separation across all birds. Acoustic correlations are bird-dependent: strong in one bird (spectral flatness $r = -0.56$), weak in others.

We do not claim this is the most efficient way to classify calls from syllables—simpler methods can do that. We propose trajectory variance as a per-vocalization measure of *developmental change*, conceptually distinct from static acoustic form.

2. Related Work

Birdsong analysis and development. Quantitative birdsong analysis has progressed from hand-crafted features [7] through supervised segmentation [8, 9] to unsupervised latent-space methods [11, 10]. These characterize vocalization structure at a given moment. Temporal dynamics of development have been studied through repertoire dating—tracking when a vocalization’s nearest neighbors were produced [12]—and through generative models linking vocal variability to circadian and developmental factors [13]. Our work is complementary: rather than classifying sounds or dating their neighbors, we ask how a vocalization would differ if produced at another developmental age.

Optimal transport in generative modeling. Flow matching [14] and rectified flow [15] learn continuous normalizing flows via conditional vector fields, and minibatch OT coupling [16] improves training stability. Our model uses OT coupling to form training pairs but predicts displacements directly rather than integrating a velocity field, resulting in a single-pass inference without ODE solvers.

Counterfactual reasoning. The question “what would this observation look like under different conditions” is central to causal inference [17] and applied in voice conversion and speech processing. To our knowledge, counterfactual age prediction has not been applied to animal vocal development.

3. Method

Our pipeline (Fig. 1) consists of three stages: a variational autoencoder compresses spectrograms into a latent space, a displacement model learns age-conditioned latent displacements, and trajectory variance is computed from counterfactual predic-

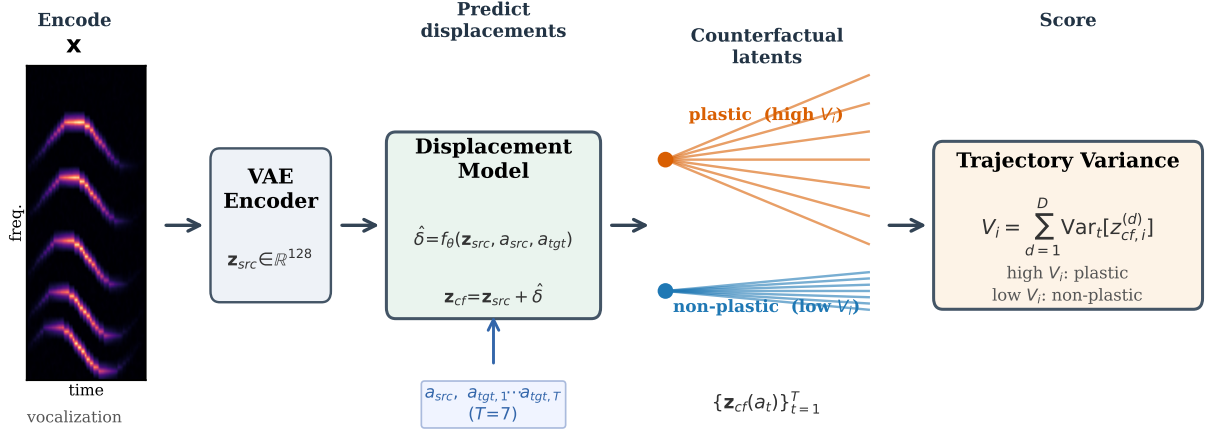


Figure 1: Pipeline overview. A VAE encodes each vocalization spectrogram into a 128-dimensional latent vector $\mathbf{z}_{\text{src}}$. The displacement model f_θ , conditioned on source and target ages, predicts a latent shift for each of $T=7$ target ages. Trajectory variance V_i is the summed per-dimension variance of the resulting counterfactual latents: learned song syllables (developmentally plastic) produce wide fans; innate calls produce narrow ones.

tions across target ages.

3.1. Dataset

Vocalizations were recorded from three juvenile male zebra finches during vocal development (40–100 dph) and segmented into individual vocalizations without type labels. Spectrograms were extracted from a pre-existing data pipeline (sample rate 32 kHz, hop length 128, 123 linear frequency bins after filtering low-frequency noise), right-padded to 100 time frames, and z-score normalized per bird. Dataset sizes: 222K (Bird A), 274K (Bird B), 183K (Bird C).

3.2. Spectrogram VAE

A convolutional variational autoencoder [18] compresses each spectrogram $\mathbf{x} \in \mathbb{R}^{123 \times 100}$ to a latent vector $\mathbf{z} \in \mathbb{R}^{128}$. The encoder applies three Conv1d layers (123→128→256→512 channels, kernel size 3, stride 2) with batch normalization and GELU activations, followed by separate linear projections to μ and $\log \sigma^2 \in \mathbb{R}^{128}$. The decoder mirrors this architecture with transposed convolutions. The loss combines masked MSE reconstruction (ignoring padding), KL divergence (weight 10^{-3}), and L_2 latent regularization (weight 10^{-4}). We use AdamW [19] (lr = 3×10^{-4}), cosine annealing, 100 epochs, batch size 128. One VAE is trained per bird.

3.3. Age-conditioned displacement model

Given encoded latents $\{(\mathbf{z}_i, a_i)\}_{i=1}^N$, where a_i denotes the developmental age (in days post-hatch) at which vocalization i was recorded, we train a model to predict the displacement $\delta = \mathbf{z}_{\text{tgt}} - \mathbf{z}_{\text{src}}$ for vocalizations resampled across ages. Because no longitudinal recordings track the same vocalization over time, training pairs are formed via mini-batch optimal transport: within each batch, we compute the pairwise L_2 cost between source- and target-age latents and solve the assignment with the Hungarian algorithm [20].

The displacement model is a 6-layer residual MLP with adaptive layer normalization (AdaLN) [21], conditioned on

source and target ages via sinusoidal embeddings. The output layer is zero-initialized so that the displacement is near-zero at initialization. The model predicts displacements in a single forward pass:

$$\mathbf{z}_{\text{cf}} = \mathbf{z}_{\text{src}} + f_\theta(\mathbf{z}_{\text{src}}, a_{\text{src}}, a_{\text{tgt}}). \quad (1)$$

Latents are z-score normalized per dimension; ages are normalized to $[\epsilon, 1-\epsilon]$ ($\epsilon = 0.05$). Training minimizes MSE on displacements. Optimization: AdamW (lr = 3×10^{-4} , weight decay 10^{-4}), cosine annealing, gradient clipping at 1.0, 200 epochs, batch size 256, with model selection on 10% held-out validation loss.

3.4. Trajectory variance

For each vocalization $\mathbf{z}_i$, we generate counterfactuals at $T=7$ evenly spaced target ages spanning the full developmental range. Trajectory variance is defined as:

$$V_i = \sum_{d=1}^D \text{Var}_{t=1}^T [z_{\text{cf},i}^{(d)}(a_t)], \quad (2)$$

where $z_{\text{cf},i}^{(d)}(a_t)$ is the d -th latent dimension of the counterfactual at target age a_t . High V_i indicates a vocalization is predicted to change substantially across development; low V_i indicates stability.

4. Experiments

All analyses are applied independently to each bird. We evaluate (1) whether trajectory variance correlates with known acoustic markers of vocal development, (2) whether it separates song syllables from calls, and (3) whether the learned displacement model outperforms simpler baselines.

4.1. Song/call labels

We label vocalizations using a temporal bout heuristic rather than acoustic clustering, avoiding circularity with the acoustic

Table 1: Pearson correlation (r) between trajectory variance and acoustic features (3K evaluation subset per bird). Spectral flatness is significant only for Bird C ($p < 10^{-10}$; Birds A, B: $p > 0.05$). All duration correlations are significant ($p < 10^{-10}$).

Feature	A	B	C
Spectral flatness	−.03	−.03	−.56
Duration	+.70	+.70	+.80

features we evaluate against. Consecutive vocalizations separated by < 200 ms gaps are grouped into bouts; vocalizations in bouts of three or more elements are labeled *song*, and all others *call*. This exploits the rapid-fire sequencing that characterizes zebra finch song practice [4], without reference to spectral content. Manual inspection of randomly sampled bouts yielded approximately 83% agreement with this heuristic, confirming that it reliably distinguishes song from calls.

4.2. Correlations with acoustic features

Table 1 reports Pearson correlations between trajectory variance and two acoustic features: spectral flatness—mathematically equivalent to Wiener entropy [7]—and vocalization duration. Duration correlates strongly across all three birds ($r = 0.70$ – 0.80). Because song syllables are generally longer than calls, this correlation means that raw trajectory variance could separate song from calls simply by reflecting duration differences. To test whether the model captures developmental structure beyond length, the song/call separation metrics in Table 2 (d_r , AUC) are computed on trajectory variance residualized on duration. Spectral flatness correlates strongly for Bird C ($r = -0.56$) but weakly for Birds A and B ($r = 0.03$). Despite the absence of spectral flatness correlations for those birds, the displacement model still separates song from calls (Table 2), suggesting trajectory variance captures developmental structure not fully reflected in static spectral descriptors.

4.3. Baseline comparison

We compare the learned displacement model against three non-parametric baselines, all operating in the same z-score normalized latent space with the same $T=7$ target ages and the same evaluation subset (3K samples per bird):

- **Gaussian OT**: Fits a Gaussian to the latent vectors at each age and applies the closed-form optimal-transport map [22] between the source- and target-age distributions. The resulting affine transform is identical for all vocalizations at a given source age, capturing only population-level drift.
- **Per-age k -NN ($k=10$)**: Mean of k nearest neighbors at each target age. Captures local structure but lacks developmental directionality.
- **Per-age OT**: Hungarian assignment between source- and target-age pools. One-to-one matching avoids the averaging of per-age k -NN but depends on pool composition.

Table 2 reports two song/call separation metrics per method: d_r , Cohen’s d on duration-residualized trajectory variance; and AUC, the area under the ROC curve for the same comparison. Figure 2 shows the corresponding distributions for the displacement model. The displacement model is the only method yielding consistently positive d_r across all three birds ($d_r = 0.29$ – 0.57 , AUC = 0.58 – 0.67), with small-to-medium

Table 2: Song/call separation: d_r = Cohen’s d on duration-residualized trajectory variance; AUC = area under ROC curve (3K subset per bird).

	d_r			AUC		
	A	B	C	A	B	C
Displacement (ours)	+.57	+.32	+.29	.67	.65	.58
Gaussian OT	+.06	−.16	+.07	.52	.44	.52
Per-age k -NN	−.05	−.07	−.03	.50	.49	.46
Per-age OT	−.19	−.41	−.24	.45	.39	.42

effect sizes. No baseline achieves consistent separation: Gaussian OT is weakly positive for two birds but negative for Bird B ($d_r = -0.16$); per-age k -NN produces near-zero d_r across all birds (AUC ≈ 0.50), indicating chance-level separation; and per-age OT shows a notable *reversal* ($d_r = -0.19$ to -0.41 , AUC = 0.39 – 0.45), assigning higher trajectory variance to calls than to songs. This reversal likely reflects the sensitivity of one-to-one matching to pool composition: when age bins are unevenly populated, stereotyped calls may be matched to distant targets, inflating their apparent variance.

5. Discussion

5.1. What trajectory variance captures

Trajectory variance from the displacement model separates song from calls across all three birds (Table 2, Fig. 2), without any acoustic feature engineering or vocalization labels. The effect sizes are small-to-medium ($d_r = 0.29$ – 0.57), consistent with the expectation that developmental plasticity is one of several factors distinguishing song from calls, rather than a perfect separator. Notably, separation is robust for Birds A and B, where spectral flatness correlations are negligible ($r = 0.03$; Table 1). This dissociation—song/call separation without spectral flatness correlation—indicates that trajectory variance is not simply a proxy for spectral flatness or any single acoustic descriptor; it captures the *predicted magnitude of developmental change*, which is a different quantity from static form. For Bird C, where strong acoustic correlations *do* emerge (spectral flatness $r = -0.56$), the model links developmental plasticity to established measures of vocal maturity [7]. The bird-to-bird variation in acoustic correlations, alongside consistent song/call separation, suggests the model captures a functional category (learned vs. innate) rather than a fixed spectral property across individuals.

5.2. Limitations

Duration correlation. Duration correlates strongly with trajectory variance across all birds ($r = 0.70$ – 0.80 ; Table 1), likely because the VAE’s fixed-length spectrograms (right-padded to 100 frames) implicitly encode vocalization length. Since song syllables are longer than calls in general, raw trajectory variance would trivially separate them. The song/call metrics (d_r , AUC) in Table 2 are therefore residualized on duration, confirming that the displacement model captures structure beyond length.

Acoustic correlations vary across birds. Spectral flatness correlations are strong only for Bird C and negligible for Birds A and B (Table 1). This may reflect differences in vocal repertoire structure or in how spectral features align with devel-

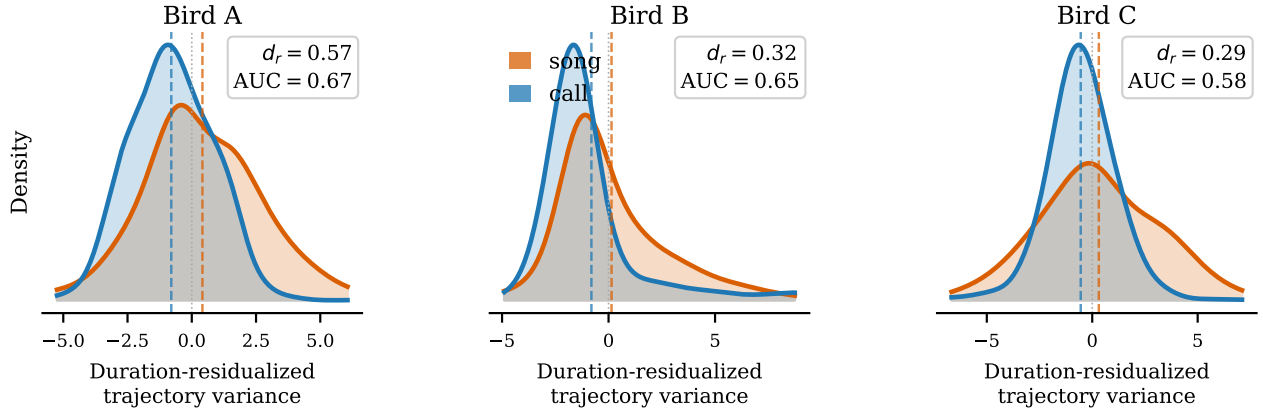


Figure 2: Distribution of duration-residualized trajectory variance for song (orange) and call (blue) vocalizations (10K samples per bird). Dashed lines mark group means. Song vocalizations receive consistently higher trajectory variance after controlling for duration, with effect sizes decreasing from Bird A ($d_r=0.57$) to Bird C ($d_r=0.29$).

opmental change across individuals. Duration correlates consistently ($r \geq 0.70$), but this is expected given that song syllables are generally longer than calls. The consistent song/call separation across all birds indicates that the *functional* distinction between learned and innate vocalizations is captured regardless.

No longitudinal ground truth. Our evidence is cross-sectional. The transport model predicts what a vocalization *would* look like at other ages based on the population, not what it *actually* became in a given individual. We cannot directly verify these counterfactual predictions.

Label heuristic. The bout-based song/call labels rely on temporal proximity, not expert annotation. This avoids circularity with acoustic features but introduces its own biases—isolated song elements would be mislabeled as calls.

Generation quality. As a sanity check, we compute Fréchet Audio Distance (FAD) [23] between decoded counterfactual spectrograms and real recordings (Griffin-Lim inversion, VGGish embeddings). Pooled FAD ranges from 0.01 to 0.06 across birds. For reference, FAD between two independent halves of the real data is 0.002–0.007; the counterfactuals are thus somewhat distinguishable from real recordings but remain on the same order of magnitude. Trajectory variance operates entirely in latent space and does not depend on the perceptual quality of decoded spectrograms.

Generalizability. We study three birds from one colony and species. Extension to other species or to human vocal development [24] remains untested.

5.3. Learned vs. nonparametric baselines

The baselines span a range of approaches: Gaussian OT applies the same affine map to all vocalizations at a given age (population-level drift only); per-age k -NN retrieves the most similar observed vocalizations at the target age; per-age OT constructs one-to-one correspondences. The learned displacement model conditions on both the source latent and the age gap, producing nonlinear, sample-specific predictions. Its advantage in song/call separation (Table 2: $d_r > 0$ for all birds, vs. near-zero or negative for all baselines) confirms that this conditioning captures developmental structure that population-level or retrieval-based methods miss. The per-age OT reversal ($d_r < 0$; calls assigned higher variance than song) is particularly informative:

one-to-one matching depends heavily on pool composition and age-bin boundaries, and does not model how a *specific* vocalization would change—only which other vocalization it most resembles. Per-age k -NN, despite capturing local latent structure, produces chance-level song/call separation ($d_r \approx 0$, $AUC \approx 0.50$), indicating that retrieval of similar vocalizations at target ages does not yield meaningful plasticity estimates.

6. Conclusion

We proposed counterfactual age displacement as a method for measuring developmental vocal plasticity without vocalization labels. Applied to three zebra finches (183K–274K vocalizations, 40–100 dph), the learned displacement model reliably separates song from calls with small-to-medium effect sizes (Cohen’s $d = 0.29$ – 0.57 , $AUC = 0.58$ – 0.67 after controlling for duration), while no nonparametric baseline achieves consistent separation. Acoustic feature correlations are bird-dependent, suggesting trajectory variance captures developmental dynamics not fully reducible to static spectral form. The approach is limited by substantial duration correlation ($r = 0.70$ – 0.80) and the absence of longitudinal ground truth, but provides a complementary perspective to static acoustic descriptions for studying vocal development.

7. Acknowledgments

8. Generative AI use disclosure

The original code, data analysis, and manuscript drafts were produced entirely by the human authors. Generative AI tools (Claude, GitHub Copilot) were used solely for editing and polishing the manuscript and formatting code. All scientific claims, experimental design, data interpretation, and final content are the sole responsibility of the human authors.

9. References

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